

MAC 233: Hall Effect Sensor Lab

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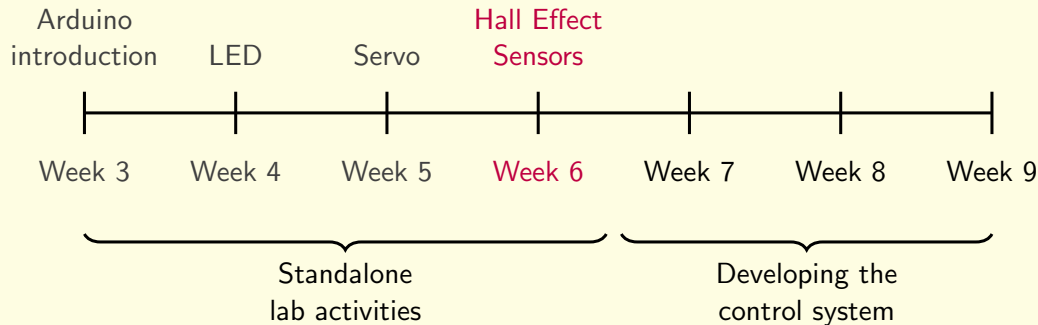
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MAC 233: **05/11/2025**



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MAC 233: LABS OVERVIEW



TODAY'S LAB: PREREQUISITES

This lab doesn't require any code from previous labs; however, it does expect you to have completed these, as common code repeated from previous labs will not be explained in detail again here.

Out of your Arduino pack, you will need: 1 Arduino, 1 A3144 hall effect sensor, 1 LED, 1 $10K\Omega$ resistor, 1 220Ω resistor, and 5 jumper cables.

TODAY'S LAB: OBJECTIVES

The objective of today's lab is to learn how to use an interrupt to record a value of a sensor whose value is changed via an external event: <https://docs.arduino.cc/language-reference/en/functions/external-interrupts/attachInterrupt/>.

READING AN INPUT: WHY WE NEED AN INTERRUPT

An input could change at any point. As such, we can't rely upon the value to change when we are at a certain point in the 'loop' function.

Instead, we have to 'interrupt' the main 'loop' function to record the value.

The 'interrupt' function can be called when the input value is 'LOW', 'CHANGE', 'RISING', or 'FALLING'.

READING AN INPUT: 'ATTACHINTERRUPT'

```
void setup() {  
    // set pin 3 to an input  
    pinMode(3, INPUT);  
    // when the input value of pin 3 falls  
    // call the function 'wheelInterrupt'  
    attachInterrupt(  
        digitalPinToInterrupt(3), wheelInterrupt, FALLING  
    );  
}  
  
void wheelInterrupt(){  
    // only called when the value associated  
    // with the pin has changed  
}
```

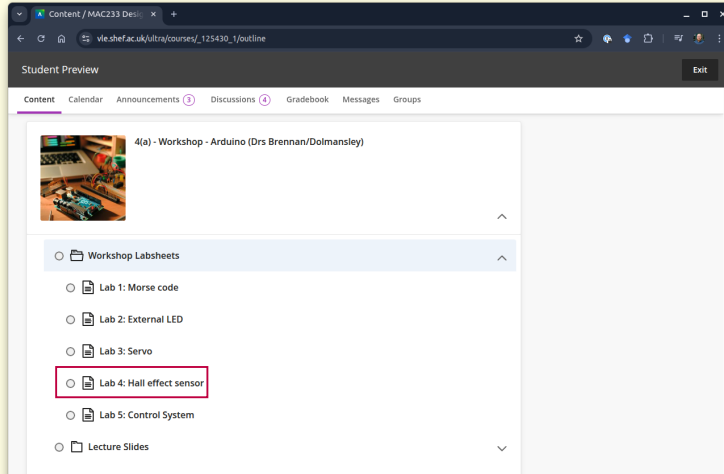
VARIABLES: 'VOLATILE'

An interrupt can change a variable at any moment. Marking it 'volatile' tells the compiler not to cache the value, so the main 'loop' always sees what the interrupt wrote.

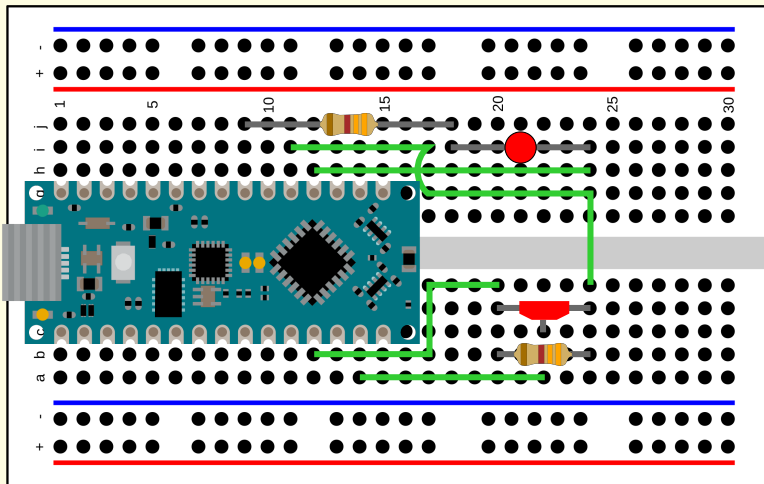
```
// 'volatile' keeps the value in RAM, so the main  
// loop always sees what the interrupt wrote.  
// Required for any variable changed by an interrupt.  
volatile unsigned int wheel_rotations_count = 0;
```

LAB 4: HALL EFFECT

Login to Blackboard: <https://vle.shef.ac.uk/>



WIRING DIAGRAM



Breadboard source: <https://freesvg.org/breadboard>, Resistor source <https://freesvg.org/resistor-330-ohm>.